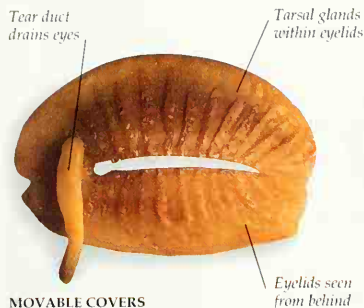
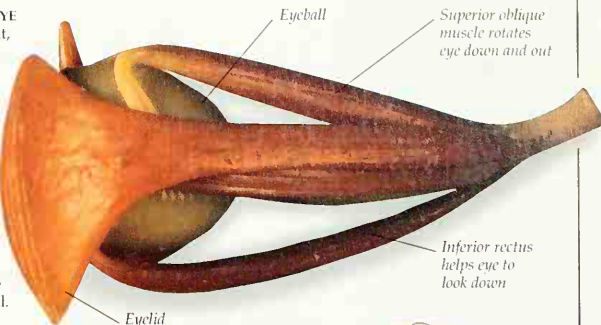




MOVABLE COVERS

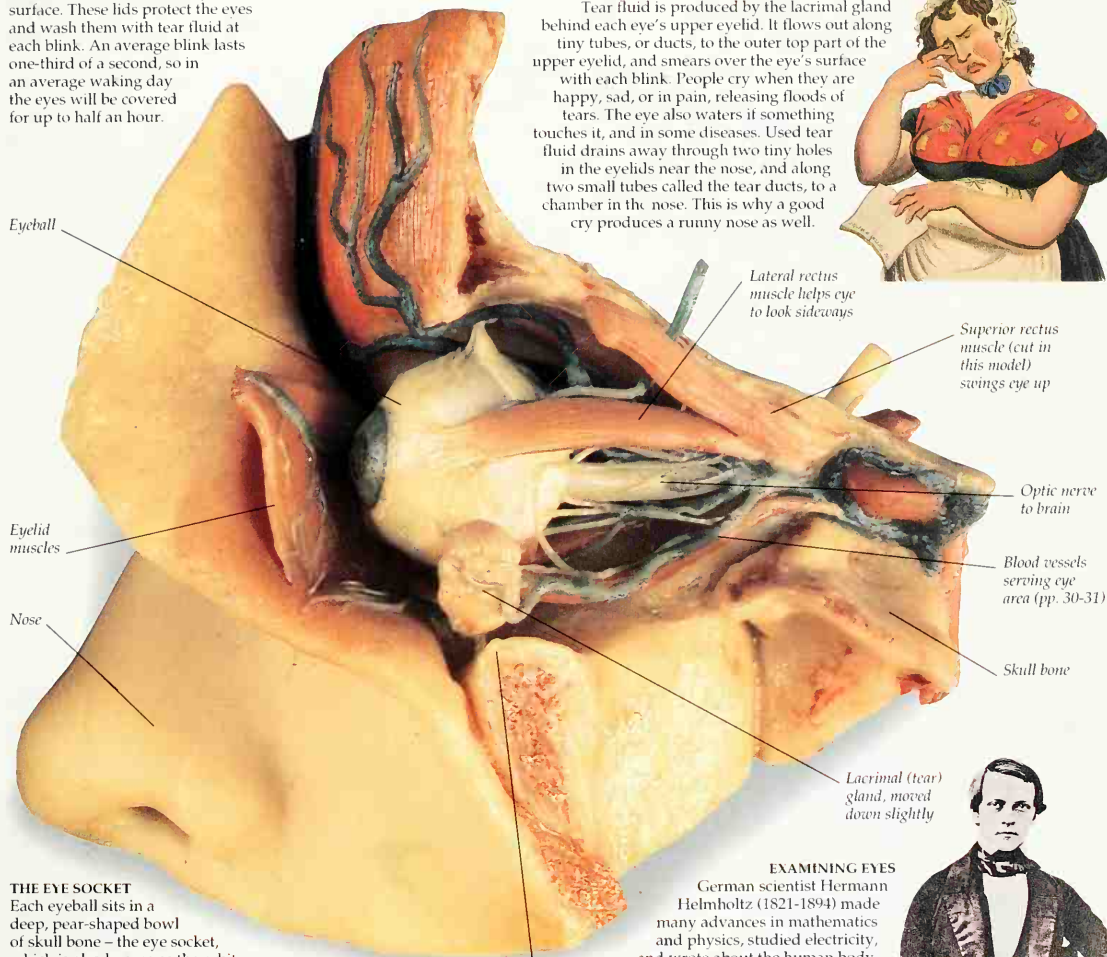
The soft, flexible eyelids are lined with the conjunctiva, which folds around and back over the eye's surface. These lids protect the eyes and wash them with tear fluid at each blink. An average blink lasts one-third of a second, so in an average waking day the eyes will be covered for up to half an hour.

MOVING THE EYE
Eyes can move right, left, up, and down. Around each eyeball are six small, slim muscles, apart from the eyelid muscles. The front end of each muscle attaches to the front sclera of the eyeball. The rear ends are anchored to the bones of the skull at the back. The muscles work as a team to make each eyeball swivel.



TEARS AND CRYING

Tear fluid is produced by the lacrimal gland behind each eye's upper eyelid. It flows out along tiny tubes, or ducts, to the outer top part of the upper eyelid, and smears over the eye's surface with each blink. People cry when they are happy, sad, or in pain, releasing floods of tears. The eye also waters if something touches it, and in some diseases. Used tear fluid drains away through two tiny holes in the eyelids near the nose, and along two small tubes called the tear ducts, to a chamber in the nose. This is why a good cry produces a runny nose as well.



THE EYE SOCKET

Each eyeball sits in a deep, pear-shaped bowl of skull bone – the eye socket, which is also known as the orbit. Between ball and bone are the muscles of the eyeball. As the eye swivels, the optic nerve angles and stretches to follow the eyeball. Both it and the eye muscles are surrounded by padded cushions of fat. The lacrimal (tear) gland is situated on the outer side of the eye.

EXAMINING EYES

German scientist Hermann Helmholtz (1821-1894) made many advances in mathematics and physics, studied electricity, and wrote about the human body, including his *Handbook of Physiological Optics* (1856-1867). He helped invent the ophthalmoscope. Doctors use this light-and-lens device for close-up examinations of the eye's interior.

